

# Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io  
 mendamar.badrall@gmail.com |  +36 70 7385 744 |  Budapest, Hungary

I'm currently a MSc Computer Science student with a BSc in Applied Mathematics and a focus on Machine Learning. I'm passionate about applying statistical techniques and algorithmic problem-solving to high-impact quantitative finance challenges.

## EDUCATION

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| <b>Budapest University of Technology and Economics (BME)</b><br>M.S. in Computer Science Engineering<br><i>Focus: Data Science &amp; Artificial Intelligence</i>  | Sep, 2025 – June, 2027<br>Budapest, Hungary     |
| <b>National University of Mongolia (NUM)</b><br>B.S. in Applied Mathematics<br>Thesis: <a href="#">Evaluating land degradation using image processing methods</a> | Sep, 2021 – June, 2025<br>Ulaanbaatar, Mongolia |

## EXPERIENCE

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| <b>Center of Mathematics Application Research Laboratory</b><br><i>Research Undergraduate Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Oct, 2023 – June, 2025 |
| <ul style="list-style-type: none"><li>– <b>Lab Leadership:</b> Selected as Undergraduate Lab Lead in final year; coordinated weekly research progress meetings and facilitated communication between students and faculty.</li><li>– <b>Infrastructure Management:</b> Managed and maintained the lab computing resources in the Linux environment.</li><li>– <b>Research:</b> Conducted research on image processing and machine learning techniques to quantify plant types from drone imagery using segmentation methods (thresholding, K-means, Moran's I autocorrelation).</li><li>– <b>Documentation:</b> Authored a comprehensive laboratory handbook and technical documentation to standardize onboarding for future researchers.</li></ul> |                        |

## PROJECTS

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### [Deep Learning Foundations and Concepts Book Figures Reproduction](#)

- Self-initiated project to reproduce key examples from C.Bishop and H.Bishop's *Deep Learning Foundations and Concepts* to gain theoretical and fundamental understanding of polynomial regression and probability theory based on **Monte Carlo Simulation**.
- Published reproducible Jupyter and Pluto.jl notebooks on GitHub, with emphasis on clear explanation, interactivity of plots, figures, and visualization.

## SKILLS

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| <b>Programming Languages:</b>  | Python, JavaScript, Julia, R, C, C++              |
| <b>ML Frameworks:</b>          | PyTorch, JAX                                      |
| <b>Markup &amp; Docs:</b>      | L <sup>A</sup> T <sub>E</sub> X                   |
| <b>Databases &amp; Tools:</b>  | Git, SQLite, MySQL                                |
| <b>Theoretical Background:</b> | Applied Mathematics, Statistics, Machine Learning |